Data-Backed Reproduction of *Ambition Is Nothing Without Focus*: A Local DISCERN–PatentView–CRSP Replication Paper

Codex local replication workflow

April 2, 2026

Abstract

This paper documents a fresh data-backed reproduction of Ghosh et al. [1] using the local datasets and code available in the workspace. The reproduction is not presented as an exact replay of the original photographic-imaging-industry sample. Instead, it is a source-faithful reconstruction of the paper's core empirical logic using the actual local data stack: DISCERN firm–patent links, WRDS/CRSP/CCM-based firm-year mapping, PatentView patent and citation files, and Compustat stocknames for classification. The pipeline was rerun on April 2, 2026. The resulting paper-comparable clean sample contains 489,193 utility patents linked to 200 GVKEYs across 31 application years. In the preferred two-way clustered Poisson specification, the interaction between recombination breadth and organizational recombination experience remains negative and statistically significant (-0.002188, $p = 0.0076$), the three-way interaction with organizational focus remains negative and statistically significant (-0.1705, $p = 0.0152$), and the interaction between organizational recombination experience and organizational focus remains positive but weakens to marginal significance (0.3219, $p = 0.0703$). The replication therefore preserves the main "experience can hurt high-breadth search unless focus compensates" pattern, while also showing that some parts of the original story are sensitive to sample tightening.

1 Introduction

Ghosh et al. [1] argues that organizations pursuing broad technological recombination face negative transfer from prior experience unless contemporaneous inventive search is sufficiently focused. The workspace already contained an executable replication pipeline built around actual local datasets rather than placeholder notes. The present document turns that pipeline into a paper-style reproduction report.

The most important scope statement is upfront. This is a data-backed reproduction of the paper's empirical design using the datasets that actually exist in the local environment. It is not an

exact claim that the original paper's proprietary or industry-specific sample has been recovered line-for-line. The current workspace uses a GVKEY-anchored patent panel built from DISCERN, PatentView, and WRDS/CRSP-linked active-year logic. That means the manuscript can report real counts, real merges, real model outputs, and real attrition from the local data, while still being honest about remaining departures from the original paper's sampling frame.

2 Original Study and Reproduction Goal

The original article is *Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D* [1]. The paper studies whether the inventive impact of high-breadth recombination depends on organizational experience with similar recombination and on the focus of the firm's contemporaneous search.

The reproduction goal here is narrower and more operational. It is to:

- rerun the full local mapping and regression pipeline from source files,
- document every major sample-construction and variable-construction decision implemented in code,
- estimate the paper-style interaction models on the actual local dataset,
- compare the raw and stricter paper-comparable variants, and
- produce a publishable PDF report from those real outputs.

3 Data Architecture

The fresh rerun executed three scripts:

- `Data_Analysis/code/run_full_replication.R`
- `Data_Analysis/code/paper_comparable_replication.R`
- `Data_Analysis/code/Regression/build_gvkey_clean_reproduction.R`

The local data inputs are summarized in Table 1. These are the files actually used in the rerun, not inferred data sources.

Table 1: Local data inputs used in the April 2, 2026 rerun

File	Source family	Use in reproduction
`firmyear_active_years_from_stocknames_plus_sp500_flag.csv`	WRDS/CRSP/CCM stocknames plus S&P 500 flag	Supplies active GVKEY-year panel and S&P 500 gate
`sp500_patents_firm_level_1986_2016_gvkey.csv`	DISCERN mapping output	Provides patent-to-GVKEY anchor for focal patents
`g_application.tsv`	PatentView based focal file	Supplies filing date and application year
`g_patent.tsv`	PatentView	Supplies grant date and patent type
`g_uspc_at_issue.tsv`	PatentView	Supplies class information for recombination variables
`g_us_patent_citation.tsv`	PatentView	Supplies backward and forward citation edges
`compustat.csv`	Compustat stocknames	Supplies company names and SIC-based sector classification

4 Firm-Year Mapping

The rerun began by rebuilding the active GVKEY-year panel. Two mapping paths were executed: Path A through CCM/CRSP and Path B through DISCERN comparison mapping. Table 2 reports the actual counts printed by the April 2, 2026 run.

Table 2: Fresh mapping-stage counts from the local rerun

Metric	Path A (CCM baseline)	Path B (DISCERN comparison)
Candidate GVKEY-years	49,631	49,631
Mappable GVKEY-years	14,320	16,281
Active GVKEY-years	14,255	13,447
Distinct GVKEY	1,269	1,243
Distinct PERMNO / adjusted PERMNO	1,327	1,235
Inactive among mappable GVKEY-years	65	2,834

Path A remains the defensible baseline because once a link exists, attrition at the active-time gate is very small. Path B maps more candidate GVKEY-years initially but loses much more of that coverage after the active-year filter. The current reproduction therefore keeps Path A as the main branch and treats Path B as a comparison path rather than as the preferred anchor.

5 Dataset Construction

The regression code makes the sample logic explicit. First, the focal patent file is merged with PatentView application dates so that application year is anchored on filing date. The code only falls back to `fyear` when the filing year is clearly impossible. In the fresh run, only 25 post-filter rows required this fallback. Second, the pipeline keeps only patents whose GVKEY-year is active

in the locally built firm-year panel. Third, recombination variables are computed from backward citations after attaching USPC main classes to cited prior-art patents. Fourth, the dependent variable is the number of forward citations received within six years after the patent's grant date, and late patents without a fully observable six-year window are dropped.

The most important sample counts are reported in Table 3. The difference between patent counts and row counts reflects ambiguous patent-to-firm assignments in the raw sample. Those are removed in the paper-comparable clean branch.

Table 3: Actual sample construction counts

Step	Count
Active GVKEY-year rows in Path A panel	14,255
Focal patent rows after active-time filter	624,096
Unique focal patents after active-time filter	619,334
Duplicate focal rows after active-time filter	4,762
Backward citation edges before class merge	10,818,457
Backward citation edges with usable class information	9,738,285
Recombinant patent rows kept (`num_classes >= 2`)	495,460
Raw regression rows after full six-year citation window	499,486
Ambiguous patent identifiers in raw regression sample	3,678
Clean paper-comparable sample rows	489,193
Clean paper-comparable unique GVKEY	200
Clean paper-comparable unique years	31

The clean sample is also economically interpretable. The top sectors are Industrials, Manufacturing & Chemicals (44.7% of patents), Technology, Electronics & IT (40.0%), Pharma/Biotech/Consumer Health (7.0%), and Oil, Gas & Natural Resources (6.1%).

6 Variable Reconstruction

The local code operationalizes the paper's constructs as follows:

- **Breadth.** Number of distinct cited USPC main classes in a patent's backward-citation set.
- **Recombination degree.** One minus the Herfindahl concentration index over cited class shares.
- **Recombination type.** Sorted concatenation of the unique cited USPC main classes.

- **Organizational recombination experience.** Log of one plus the cumulative number of prior patents by the same GVKEY that use the same recombination type.
- **Organizational focus.** Herfindahl index over recombination-type shares within a GVKEY-year.
- **Field recombination experience.** Log of one plus the cumulative number of prior patents in the sample using the same recombination type.
- **Inventive impact.** Number of forward citations received within six years after grant date.

The core formulas used by the script are:

$$\text{RecombDegree}_p = 1 - \sum_{c \in C_p} s_{pc}^2, \tag{1}$$

$$\text{OrgFocus}_{it} = \sum_{r \in R_{it}} s_{irt}^2, \tag{2}$$

$$\text{InventiveImpact}_p = \sum_q \mathbf{1}\{q \text{ cites } p \text{ and } g_q \in [g_p, g_p + 6 \text{ years}]\}. \tag{3}$$

The estimating equation is a fixed-effects Poisson model:

$$\mathrm{E}[Y_{pit} \mid X_{pit}, \alpha_i, \delta_t] = \exp\big(\alpha_i + \delta_t + \beta_1 \text{Breadth}_{pit} + \beta_2 \text{OrgExp}_{pit} + \beta_3 \text{Focus}_{it} + \beta_4 \text{Breadth}_{pit}\text{OrgExp}_{pit} + \beta_5 \text{Breadt} \tag{4}$$

where α_i are GVKEY fixed effects, δ_t are application-year fixed effects, and Z_{pit} contains citation, recombination, scale, and age controls. In practice, `firm_age` is dropped by collinearity in the fixed-effects regressions.

7 Main Results

The preferred specification for the paper-style reproduction is the clean sample that keeps only S&P 500 firm-years, utility patents, and unambiguous patent-to-firm assignments. Table 4 reports the main two-way clustered Poisson estimates from that sample.

Three features stand out. First, the breadth-by-experience interaction remains negative and statistically significant. Second, the three-way interaction also remains negative and statistically significant. Third, the experience-by-focus interaction is still positive, but it weakens relative to the looser raw sample and no longer clears the 5% threshold under two-way clustering.

Table 4: Preferred clean-sample baseline: Poisson with GVKEY and application-year fixed effects

Term	Estimate	p-value
Breadth	0.0030	0.8477
Organizational recombination experience	0.0706	0.0461
Organizational focus	-0.3609	0.3373
Backward citations	0.001839	0.0268
Recombination degree	-0.4546	0.0187
Field recombination experience	-0.1139	0.0127
Firm patent stock up to $t-1$	-0.1661	0.0477
Breadth × org. experience	-0.002188	0.0076
Breadth × focus	-0.0050	0.9622
Org. experience × focus	0.3219	0.0703
Breadth × org. experience × focus	-0.1705	0.0152
Observations used by model	489,183	
Pseudo-R^2	0.2400	

8 Sensitivity to Sample Tightening

Because the workspace contains multiple defensible dataset variants, the cleanest way to assess robustness is to compare the headline interactions across those variants. Table 5 reports the fresh two-way clustered estimates.

Table 5: Headline interaction terms across dataset variants

Term	Raw current sample	S&P 500 only	Clean drop-ambiguous sample
Breadth × org. experience	-0.002179	-0.002190	-0.002188
p-value	0.0053	0.0045	0.0076
Org. experience × focus	0.3333	0.3286	0.3219
p-value	0.0254	0.0564	0.0703
Breadth × org. experience × focus	-0.1624	-0.1651	-0.1705
p-value	0.0254	0.0209	0.0152

This comparison matters for interpretation. The negative breadth-by-experience interaction is remarkably stable across variants. The negative three-way interaction is also stable. By contrast, the positive experience-by-focus interaction is the result most exposed to sample tightening. That pattern implies the workspace's main finding is not that every original coefficient is mechanically replicated, but that the key negative-transfer logic survives stricter cleaning.

9 Additional Robustness

The clean reproduction script also estimates year-relative high-impact logits. In the top-5% by year logit, the breadth-by-experience interaction remains strongly negative (-0.0118, $p < 0.001$), while the three-way interaction is not statistically significant. In the top-1% by year logit, the breadth-by-experience interaction remains negative (-0.0132, $p = 0.0184$), while the focus-related interaction terms are again not statistically reliable. These checks suggest that the negative-transfer mechanism is more stable than the compensation-through-focus term when the dependent variable is sharpened to extreme-impact indicators.

10 What Was Actually Reproduced

This reproduction improves substantially over a note-only summary because it uses the actual local datasets and reruns the code end-to-end. The following pieces are genuinely data-backed in this report:

- firm-year mapping from local Path A and Path B scripts,
- active-time filtering at the GVKEY-year level,
- patent-level recombination-variable construction from PatentView class and citation files,
- six-year forward-citation dependent variable construction,
- fixed-effects Poisson estimation and clustered inference,
- stricter paper-comparable sample restrictions, and
- model-fit and sector-composition summaries.

At the same time, the manuscript does not overclaim. The current workspace does not transparently recover the original article's full photographic-imaging-industry sample, and the local reproduction uses a broader GVKEY-anchored patent panel filtered through S&P 500 active years. For that reason, the correct label is a *data-backed paper-comparable reproduction*, not a proven exact line-by-line replication of the original published dataset.

11 Conclusion

The user's correction was substantively important: a real replication needs actual data, not just methodological prose. Once the local DISCERN–PatentView–WRDS stack is used directly, the project yields a much sharper conclusion. The April 2, 2026 rerun preserves the core negative

interaction between recombination breadth and organizational recombination experience, and it preserves the negative three-way interaction with focus in both the raw and clean paper-comparable samples. What weakens is the positive interaction between organizational recombination experience and organizational focus. The clean-sample reproduction therefore supports the paper's negative-transfer logic more strongly than it supports the full compensatory mechanism in every specification.

A Local rerun provenance

This PDF was written from outputs generated by the following local reruns on April 2, 2026:

- `Data_Analysis/code/run_full_replication.R`
- `Data_Analysis/code/paper_comparable_replication.R`
- `Data_Analysis/code/Regression/build_gvkey_clean_reproduction.R`

Key output files used in the manuscript include:

- `regression_summary_gvkey.txt`
- `paper_comparable_step_counts.csv`
- `paper_comparable_focus_table.csv`
- `dataset_situation_diagnostics.csv`
- `gvkey_clean_reproduction_sample_summary.csv`
- `gvkey_clean_reproduction_model_fit.csv`
- `gvkey_clean_reproduction_model_coefficients.csv`

References

[1] Anindya Ghosh, Xavier Martin, Johannes M. Pennings, and Filippo Carlo Wezel. Ambition is nothing without focus: Compensating for negative transfer of experience in r&d. *Organization Science*, 25(2):572–590, 2014. doi: 10.1287/orsc.2013.0845.